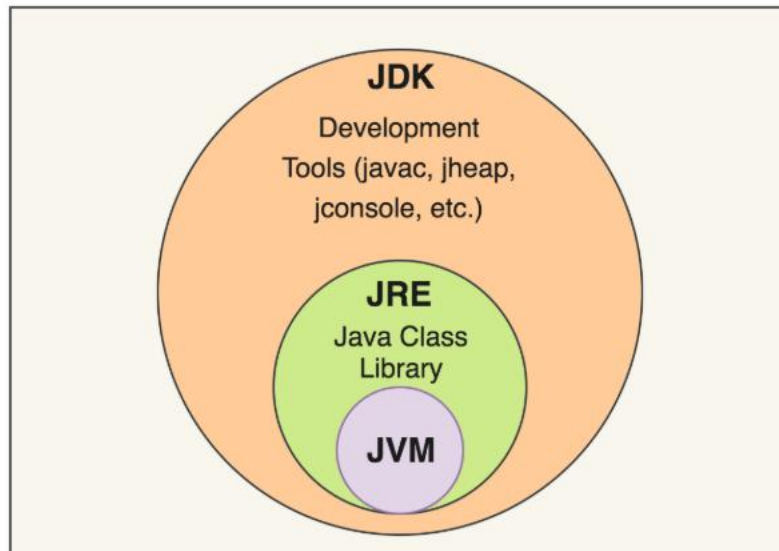


## Java Development Kit (JDK)

The **Java Development Kit (JDK)** is a software package that provides the tools required to **develop, compile, debug, and run Java applications**.

**JDK = JRE + Development Tools**



**What does JDK contain?**

### 1. JRE (Java Runtime Environment)

- Required to run Java programs.
- Contains the JVM and Java libraries.

### 2. JVM (Java Virtual Machine)

- Executes Java byte code.
- Makes Java platform-independent.

### 3. Development Tools

Some important tools included in the JDK are:

| Tool    | Purpose                                     |
|---------|---------------------------------------------|
| javac   | Compiles .java files into .class files      |
| java    | Runs Java programs                          |
| javadoc | Generates documentation                     |
| jar     | Creates JAR (Java Archive) files            |
| jdb     | Java debugger                               |
| javap   | Displays information about compiled classes |
| jshell  | Interactive Java shell (Java 9+)            |

## Java Program Execution

Step 1: Write the program

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello Java");  
    }  
}
```

Main.java

Step 2: Compile using JDK

javac Main.java

Output: Main.class

Step 3: Run using JVM

java Main

Output: Hello Java